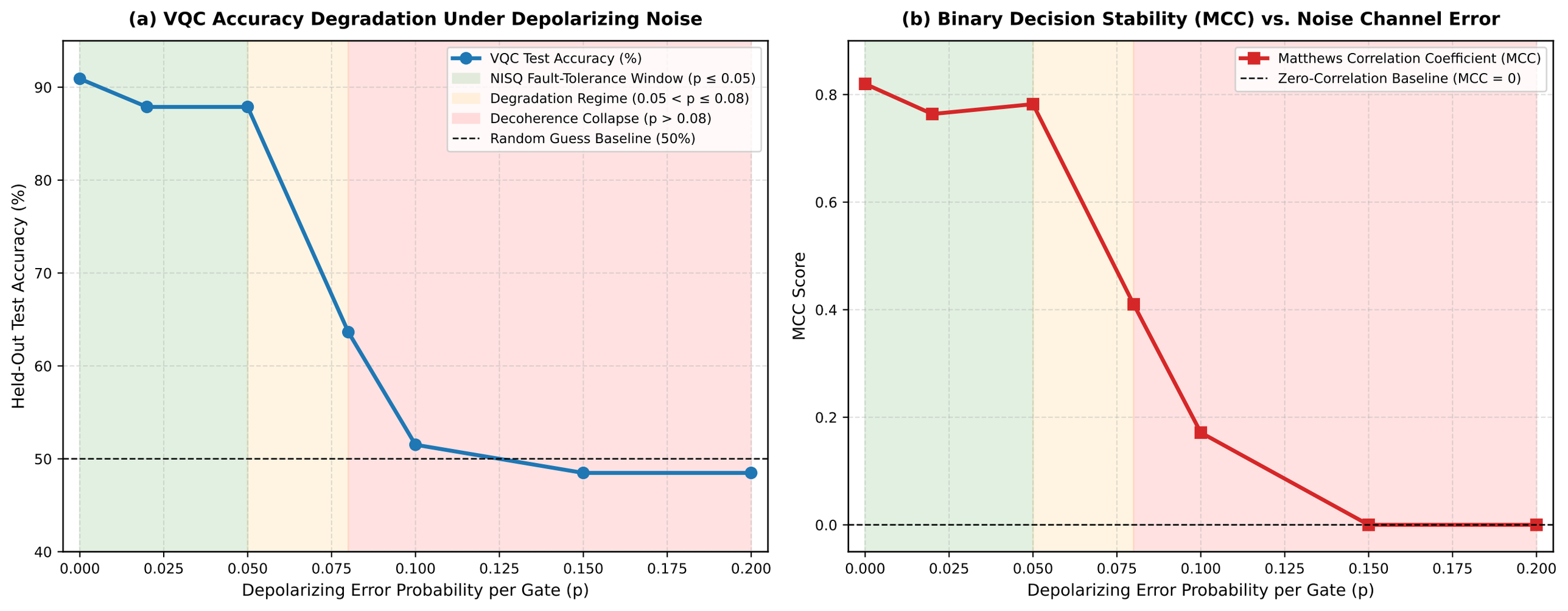
(a) VQC Accuracy Degradation Under Depolarizing Noise
VQC Test Accuracy (%)
NISQ Fault-Tolerance Window (p ≤ 0.05)
Degradation Regime (0.05 < p ≤ 0.08)
Decoherence Collapse (p > 0.08)
Random Guess Baseline (50%)
Held-Out Test Accuracy (%)
90
80
70
60
50
40
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Depolarizing Error Probability per Gate (p)
(b) Binary Decision Stability (MCC) vs. Noise Channel Error
Matthews Correlation Coefficient (MCC)
Zero-Correlation Baseline (MCC = 0)
MCC Score
0.8
0.6
0.4
0.2
0.0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Depolarizing Error Probability per Gate (p)